

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	5

Case Study Topic: Snapdeal Product Catalog Sorting using Timsort (Hybrid Stable Sorting Algorithm)

Word Done Summary:

In this case study, we analyzed **Timsort**, a hybrid sorting algorithm used in Python and Java (Arrays.sort() for Objects). Timsort is specifically designed to efficiently sort **partially sorted data**, which is common in real-world applications such as Snapdeal's product catalog management. The catalog receives daily product updates from sellers, where most new products are appended at the end, creating naturally sorted segments called **runs**.

The given dataset:

[3, 7, 11, 5, 4, 6, 9, 12, 1, 2, 8, 10, 13, 14]

It was analyzed using Timsort's run detection mechanism.

Implementation Details:

1. Natural run detection.
2. Reversal of descending runs.
3. Stable merging of runs.
4. Insertion sort for small runs.
5. Final sorted output generation.
6. Comparison with Merge Sort, Quick Sort, and Heap Sort.

Result:

Input

[3, 7, 11, 5, 4, 6, 9, 12, 1, 2, 8, 10, 13, 14]

Detected Runs

Run1 = [3,7,11]

Run2 = [4,5]

Run3 = [6,9,12]

Run4 = [1,2,8,10,13,14]

Final Sorted Array

[1,2,3,4,5,6,7,8,9,10,11,12,13,14]

Timsort combines the advantages of Insertion Sort and Merge Sort. By detecting already sorted runs and merging them efficiently, it achieves excellent performance on real-world datasets. This makes it the preferred sorting algorithm in many programming languages and large-scale applications such as e-commerce product catalogs.

Output:

```
Original Array:
3 7 11 5 4 6 9 12 1 2 8 10 13 14

Sorted Array:
1 2 3 4 5 6 7 8 9 10 11 12 13 14
```

GitHub Link:

<https://github.com/saanvikompelli-cmd/DSA-2-Case-Studies>

Student Signature	Faculty Signature
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